Table 1: Vehicle parameters and model variables of the diagonal dual-steer AGV.

Category	Symbol	Description	Value	Unit / Note
Vehicle parameter	m	Vehicle mass	200.0000	kg
Vehicle parameter	I_z	Yaw moment of inertia	16.6700	$\mathrm{kg\,m^2}$
Vehicle parameter	L	Wheelbase-related length	2.0000	m
Vehicle parameter	W	Track width	0.8000	m
Vehicle parameter	h_{cg}	Center-of-gravity height	0.5000	m
Vehicle parameter	r_w	Wheel radius	0.1500	m
Vehicle parameter	μ	Tire-road friction coefficient	0.8000	–
Vehicle parameter	C_{rr}	Rolling resistance coefficient	0.0150	–
Vehicle parameter	$C_d A_f$	Drag coefficient-area product	0.5000	$\mathrm{m^2}$
Vehicle parameter	g	Gravitational acceleration	9.8100	$\mathrm{m/s^2}$
State variable	X	Global x-position	state variable	m
State variable	Y	Global y-position	state variable	m
State variable	ψ	Heading angle	state variable	rad
State variable	v	Longitudinal velocity	state variable	m/s
State variable	ω	Yaw rate	state variable	rad/s
State variable	δ_{lf}	LF steering angle	state variable	rad
State variable	δ_{rr}	RR steering angle	state variable	rad
State variable	β	Sideslip angle	state variable	rad
Control input	F_{cmd}	Driving force command	control input	N
Control input	ω_{cmd}	Yaw-rate command	control input	rad/s
Slope variable	θ	True road slope angle	model variable	rad
Slope variable	θ_k^{sch}	Scheduled slope used by LPV-MPC	scheduler output	rad
Slope variable	$\hat{\theta}_k$	Slope-related temporal estimate	estimator output	rad

Table 2: LPV-MPC settings and constraints used in the closed-loop simulations.

Item	Symbol	Value	Unit / Description
Sampling time	T_s	0.0100	s
Prediction horizon	N_p	160	1.60 s
Control horizon	N_c	60	0.60 s
Tracking weight	Q	diag(15.293, 28.737, 5.076, 2.9918)	$e_y, e_\psi, e_v, e_\omega$
Input weight	R	diag(0.001, 0.001)	F_{cmd}, ω_{cmd}
Input-increment weight	R_Δ	diag(0.01, 0.01)	$\Delta F_{cmd}, \Delta\omega_{cmd}$
Input constraint	F_{cmd}	$[-600, 600]$	N
Input constraint	ω_{cmd}	$[-1.2, 1.2]$	rad/s
Input-increment constraint	ΔF_{cmd}	$[-400, 400]$	N/step
Input-increment constraint	$\Delta\omega_{cmd}$	$[-0.9, 0.9]$	(rad/s)/step
Output soft constraint	e_y	$[-1.0, 1.0]$	m, ECR 3×10^3
Output soft constraint	e_ψ	$[-0.5, 0.5]$	rad, ECR 3×10^3
Implementation	Solver	MATLAB MPC Toolbox	Adaptive MPC with online model update

Table 3: Temporal perception input window and multi-task output definition.

Component	Symbol / Setting	Value	Description
Input window	L	128	Historical samples
Input dimension	F	19	Observable features
Sampling time	T_s	0.0100	s
Window duration	LT_s	1.28	s
Input tensor	Z_k	$\mathbb{R}^{128\times 19}$	Train-set-normalized window
Normalization	Scaler	fit_train_only_apply_val_test_online	Fit on training split only
Split policy	Split	run_level_no_window_leakage	Run-level split
Feature categories	Inputs	19 named features	Acceleration, yaw rate, steering, wheel
Main-condition task	Head 1	flat / stall / slope	Auxiliary classification output
Steering-direction task	Head 2	right / straight / left	Auxiliary classification output
Slope regression task	Head 3	$\hat{\theta}_k$	Processed by $S(\cdot)$ before scheduling
Control interface	θ_k^{sch}	LPV-MPC scheduler input	Network does not directly output F_{cmd}

Table 4: Model configurations of GRU, TCN, and ModernTCN.

Model	Temporal encoder	Main configuration	Kernel	Ca
GRU	2-layer GRU	hidden=96; pooling=last_mean_inputstats; dropout=0.20	–	no
TCN	Causal dilated Conv1D	filters=96; blocks=6; dropout=0.15	3	ca
ModernTCN	Depthwise large-kernel Conv1D	channels=64; blocks=5; expansion=2; dropout=0.15	31	sa

Table 5: Baseline controllers and estimators used for closed-loop comparison.

Category	Method	Scheduled slope source	Role
No-slope baseline	LPV-MPC theta0	$\theta_k^{sch} = 0$	Nominal zero-slope scheduling
Sensor baseline	LPV-MPC IMU theta	$\theta_k^{sch} = \theta_k^{imu}$	Simplified sensor-based scheduling
Oracle baseline	LPV-MPC oracle theta	$\theta_k^{sch} = \theta_k^{true}$	True-slope scheduling upper bound
Learning baseline	GRU	$\theta_k^{sch} = S(\hat{\theta}_k^{GRU})$	Recurrent temporal estimator
Learning baseline	TCN	$\theta_k^{sch} = S(\hat{\theta}_k^{TCN})$	Conventional convolutional estimator
Proposed method	ModernTCN	$\theta_k^{sch} = S(\hat{\theta}_k^{ModernTCN})$	Multi-task temporal scheduling estimator
Ablation	Causal ModernTCN	$\theta_k^{sch} = S(\hat{\theta}_k^{causal})$	Offline–closed-loop mismatch analysis

Table 6: Main closed-loop results on the factory logistics showcase path.

Controller	e_y RMSE (m)	e_ψ RMSE (rad)	XY RMSE (m)	$J_{\Delta u}$	Violation rate	θ^{sch} MAE (d
ModernTCN	0.0246	0.0438	0.8036	0.5991	0	0.9
GRU	0.0623	0.1714	1.6641	7.0298	0	0.3
TCN	0.0364	0.1061	1.3538	235.6737	5.277×10^{-5}	1.7
LPV-MPC theta0	26.2527	1.6517	32.0981	285.9290	0.1040	3.1
LPV-MPC IMU theta	26.4718	1.6155	32.1210	285.8180	0.1040	2.9
LPV-MPC oracle theta	0.0236	0.0315	0.5602	0.0609	0	0.0

Table 7: Aggregate closed-loop results over three routes.

Controller	Paths	e_y RMSE mean	e_ψ RMSE mean	XY RMSE mean	$J_{\Delta u}$ mean	Overall rank me
ModernTCN	3	0.0319	0.0321	0.4889	13.3133	1.00
LPV-MPC oracle theta	3	0.0347	0.0294	0.3951	0.4386	2.00
GRU	3	0.0483	0.0851	0.9358	3.3528	3.33
TCN	3	0.0489	0.0814	1.0695	313.9197	3.66
LPV-MPC IMU theta	3	8.8930	0.6669	12.1776	579.7648	5.00
LPV-MPC theta0	3	8.8276	0.6891	12.2527	644.0558	6.00

Table 8: Robustness aggregate results under disturbance levels.

Level	Controller	Cases	e_y RMSE mean	XY RMSE mean	$J_{\Delta u}$ mean	Overall rank mean
0	ModernTCN	2	0.0355	0.3316	19.6704	1.0000
0	GRU	2	0.0413	0.5716	1.5143	2.0000
0	TCN	2	0.0552	0.9273	353.0426	3.0000
1	ModernTCN	2	0.0618	1.2880	205.7775	1.0000
1	GRU	2	0.0765	1.3874	307.3123	2.5000
1	TCN	2	0.0686	1.4244	669.0753	2.5000
2	ModernTCN	2	0.0819	1.6587	709.7593	1.0000
2	GRU	2	0.1170	1.8842	292.1807	2.0000
2	TCN	2	0.0957	1.8597	1101.8056	3.0000

Table 9: Offline perception and causal ModernTCN ablation results.

Metric group	Metric	Direction	Default ModernTCN	Causal ModernTCN	Causal / Default
Offline perception	Main acc. (%)	$\uparrow$	98.07	97.83	0.9975
Offline perception	Turn acc. (%)	$\uparrow$	90.22	90.06	0.9982
Offline perception	Trans.-turn acc. (%)	$\uparrow$	77.57	79.07	1.0193
Offline perception	Slope MAE (deg)	$\downarrow$	0.2519	0.2507	0.9955
Closed loop	e_y RMSE (m)	$\downarrow$	0.0246	23.1318	940.7977
Closed loop	e_ψ RMSE (rad)	$\downarrow$	0.0438	1.2734	29.1035
Closed loop	XY RMSE (m)	$\downarrow$	0.8036	27.4839	34.2030
Closed loop	$J_{\Delta u}$	$\downarrow$	0.5991	268.8678	448.8038

Table 10: Computational feasibility check.

Component	Metric	Mean	P95	Max	Unit	Platform
Temporal estimator	ModernTCN ONNXRuntime core inference	0.3804	0.4507	1.8839	ms/step	Windows-
MPC solver	LPV-MPC solve time	0.0268	0.0416	2.1766	ms/step	–
Core cycle	ONNXRuntime inference + MPC solve	0.4072	0.4923	4.0605	ms/step	p95 sum o
MATLAB wrapper	MATLAB replay update + predict (steady)	17.1346	28.3888	42.4473	ms/step	–
Simulation wrapper	Desktop Simulink wall time	29.3024	30.0940	30.0940	ms/step	desktop si
Sampling reference	Control sampling period	10.0000	–	–	ms	Simulatio